

PART A

# Fundamental principles

# Sport and exercise medicine: the team approach

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*You may have the greatest bunch of individual stars in the world,  
but if they don't play together, the club won't be worth a dime.*

Babe Ruth (1895-1948)

Sport and exercise medicine (SEM) is evolving as a specialist medical discipline that includes a variety of tasks and responsibilities.<sup>1-7</sup> These include:

- injury and illness prevention
- injury diagnosis, treatment and rehabilitation
- management of medical problems
- performance enhancement through training
- nutrition and psychology
- exercise prescription in health and in chronic disease states
- exercise prescription in special subpopulations
- medical care of sporting teams and events
- medical care in situations of altered physiology, such as at altitude, environmental extremes, or at depth
- dealing with ethical issues, such as the problem of drug abuse in sport.

SEM has been defined as the scope of medical practice that focuses on:

1. prevention, diagnosis treatment and rehabilitation of injuries that occur during physical activity
2. prevention, diagnosis, and management of medical conditions that occur during or after physical activity
3. promotion and implementation of regular physical activity in the prevention, treatment and rehabilitation of chronic diseases of lifestyle.<sup>8</sup>

Because of the growing breadth of content, SEM traditionally lends itself to being practised by a multidisciplinary team of professionals with specialised skills who provide optimal care for the athlete and improve each other's knowledge and skills. The sporting adage that a 'champion team' would always beat a 'team of champions' applies to sport and exercise medicine. This team approach

can be implemented in a multidisciplinary SEM clinic or by individual practitioners of different disciplines collaborating by cross-referral. However, the real-world application of this multiskilled team approach poses significant challenges—some of which are eloquently dealt with by the specialist SEM physician. SEM is now an official medical specialty in many countries and this has significantly changed the level of care to athletes and also patients suffering from chronic disease.

## THE SEM TEAM

The most appropriate sports medicine team depends on the setting. Clubs and sporting bodies increasingly employ specialist SEM physicians, with the primary role being the comprehensive health management of the athlete to facilitate optimal performance. This includes the diagnosis and treatment of injuries and illnesses associated with exercise to ultimately improve performance. In an isolated rural community the sports medicine team may consist of a family physician or a physiotherapist/physical therapist alone. In a populous city, the team may consist of a number of clinicians and sports scientists working together and may include:

- specialist SEM physician
- family physician
- physiotherapist/physical therapist
- soft tissue therapist
- exercise specialist for exercise prescription
- other medical specialists with an interest in SEM
- orthopaedic surgeon, rheumatologist, radiologist, cardiologist, etc.
- podiatrist
- dietitian/nutritionist

- psychologist
- sports trainer/athletic trainer
- other professionals such as osteopaths, chiropractors, exercise physiologists, biomechanists, nurses, occupational therapists, orthotists, optometrists
- coach
- fitness adviser.

In the Olympic polyclinic, an institution that aims to serve all 10 000 athletes at the games, the sports medicine team includes 160 practitioners (Table 1.1).

### Multiskilling, roles, responsibilities and communication

There are, of course, distinct differences in roles and responsibilities when clinicians are employed by sporting organisations or clubs as opposed to providing care in the form of a once-off expert opinion for a specific athlete's injury or illness. Equally, medical teams contracted by the

local organising committees (LOCs) of smaller or major events have very specific roles and skills depending on the size of the event. Multiskilling, where practitioners in the team each develop skills in a particular area of sports medicine, is important but may also pose challenges in areas of considerable amount of overlap between the different practitioners. The key is always effective teamwork with clear roles and responsibilities, leadership and communication. This 'multiskilling' might be of use if the practitioner is geographically isolated or is travelling alone with sporting teams.

The concept of effective teamwork is best illustrated by example. When an athlete presents with an overuse injury of the lower limb, the specialist SEM (or other physician) will have the diagnostic responsibility but the sport-specific therapist, podiatrist or biomechanist might have a better knowledge of clinical biomechanical assessment, the functional relationship between abnormal biomechanics and the development of the injury and how to correct any biomechanical cause. However, it is essential that all the practitioners have a basic understanding of lower limb biomechanics and are able to perform a clinical assessment.

Similarly, for the athlete who presents complaining of excessive fatigue and poor performance, the dietitian is best able to assess the nutritional state of the athlete and determine if a nutritional deficiency is responsible for the patient's symptoms. However, other practitioners such as the SEM physician, physiotherapist/physical therapist or trainer must also be aware of the possibility of nutritional deficiency as a cause of tiredness and be able to perform a brief nutritional assessment.

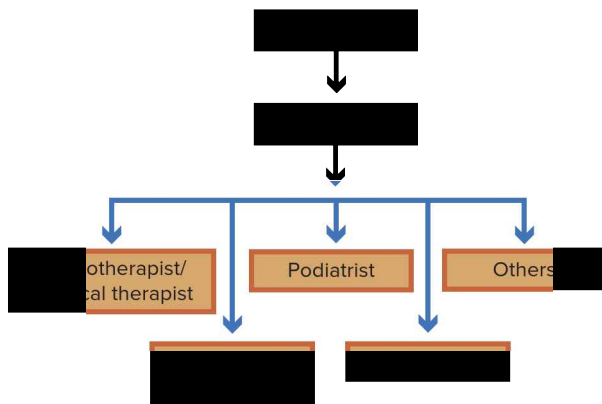
### The sport and exercise medicine model

The model for the delivery of effective healthcare is changing with an increased focus on individualised medical care and case managing. The traditional medical model (Fig. 1.1) has the physician as the primary contact practitioner with subsequent referral to other medical and paramedical practitioners. The early SEM model (Fig. 1.2) is different. The athlete's primary professional contact is often with a physiotherapist/physical therapist; however, it is just as likely to be a trainer, SEM physician or soft tissue therapist. It is essential that all practitioners in the healthcare team understand their own strengths and limitations and are aware of who else can improve management of the patient.

This improved SEM model recognises the multidisciplinary nature of the athlete's 'primary professional contact' but still emphasises the reductionist approach, with each discipline potentially operating

**Table 1.1 The clinical team structure at the London 2012 Olympic Games polyclinic**

| Administration/organisation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Chief Medical Officer</li> <li>• Deputy Chief Medical Officer and Chief Athlete Services (sports physician)</li> <li>• Director of Clinical Services—polyclinic (sports physician)</li> <li>• Director of Nursing</li> <li>• Director of Physiotherapy/Physical therapy</li> <li>• Director of Remedial Massage</li> <li>• Director of Podiatric Services</li> <li>• Director of Dental Services</li> <li>• Director of Emergency Services</li> </ul>                                                                                        |
| Consulting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>• Medical practitioners: sports physicians; orthopaedic surgeons; general practitioners; rehabilitation specialists; emergency medicine specialists; ear, nose and throat specialists; gynaecologists; dermatologists; ophthalmologists; ophthalmic surgeons; radiologists; amputee clinic physicians; spinal clinic physicians</li> <li>• Physiotherapists/Physical therapists</li> <li>• Soft tissue therapists</li> <li>• Podiatrists</li> <li>• Optometrists</li> <li>• Pharmacists</li> <li>• Dentists</li> <li>• Interpreters</li> </ul> |



**Figure 1.1** The traditional medical model

in its own specialist silo with little focus on holistic athlete health management, effective communication integration and understanding to facilitate decision making. The athlete and coach are often ill equipped to integrate the different contributions in these settings of increasing complexity and multispecialists. They often rely on expert case managers with a very good sport-specific understanding (in big teams the specialist SEM physician) to provide 'health leadership', effectively and efficiently communicating and integrating the contributions of all the key role players in the so-called *integrated performance health management and coaching model*.<sup>9</sup> This new model emphasises the importance of an integrated approach in communication and management to athlete health problems. The focus is on operational integration of the health management and coaching to improve performance with one accountable 'case manager'.

## THE CHALLENGES OF MANAGEMENT

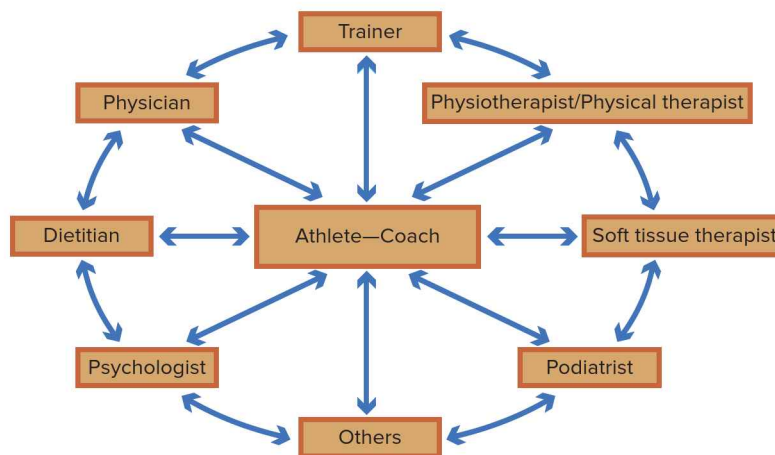
The secret of success in SEM is to take a broad view of the patient and his or her problem. The narrow view may provide short-term amelioration of symptoms but will ultimately lead to failure. Examples of a narrow view may include a runner who presents with shin pain, is diagnosed as having a stress fracture of the tibia and is treated with rest until pain-free.

Although it is likely that in the short term the athlete will improve and return to activity there remains a high likelihood of recurrence of the problem on resumption of activity. The clinician must always ask 'Why has this injury/illness occurred?' The cause may be obvious, for example, a recent sudden doubling of training load, or it may be subtle and, in many cases, multifactorial.

The greatest challenge of SEM is to identify and correct the cause of the injury/illness. The runner with shin pain arising from a stress fracture may have abnormal biomechanics, inappropriate footwear, had a change in the training surface, or a change in the quantity or quality of training. In medicine, there are two main challenges: diagnosis and treatment. In SEM it is necessary to diagnose both the problem and the cause. Treatment then needs to be focused on both these areas.

## Diagnosis

Every attempt should be made to diagnose the precise anatomical, pathological and functional cause of the presenting problem. Knowledge of anatomy (especially surface anatomy) and an understanding of the pathological and functional processes likely to occur in athletes often permits a precise diagnosis. Thus, instead of using a purely descriptive term such as 'shin splints', the practitioner



**Figure 1.2** The SEM model. In professional sport the player's agent also features prominently in athlete-coach interaction

should attempt to diagnose which of the three underlying causes it could be—stress fracture, chronic compartment syndrome or periostitis—and use the specific term. Accurate diagnosis guides precise treatment.

Some clinical situations do not allow a precise anatomical and pathological diagnosis. For example, in many cases of low back pain, it is clinically impossible to differentiate between potential sites of pathology. In situations such as these it is necessary to establish a functional diagnosis, monitor symptoms and signs through careful clinical assessment and correct any abnormalities present (e.g. hypomobility) using appropriate treatment techniques.

As mentioned, diagnosis of the presenting problem should be followed by diagnosis of the cause of the problem. American orthopaedic surgeon Ben Kibler coined the term 'victim' for the presenting problem and 'culprit' for the cause.<sup>10</sup> Diagnosis of the cause often requires a good understanding of biomechanics, technique, training, nutrition and psychology. Just as there may be more than one pathological process contributing to the patient's symptoms, a combination of factors may cause the problem.

As with any branch of medicine, diagnosis depends on careful clinical assessment, which consists of obtaining a history, physical examination and investigations. The most important of these is undoubtedly the history but, unfortunately, this is often neglected. It is essential that the sports clinician be a good listener and develop skills that enable him or her to elicit the appropriate information from the athlete. Once the history has been taken, an examination can be performed. It is essential to develop examination routines for each joint or region and to include in the examination an assessment of any potential causes.

Investigations should be regarded as an adjunct to, rather than a substitute for, adequate history and examination.<sup>11</sup> The investigation must be appropriate to the athlete's problem, provide additional information and should only be performed if it will affect the diagnosis and/or treatment.

## Treatment

Ideally, treatment has at least three components: discussing the planned treatment with the athlete, coach and key role players (also in the context of the immediate and future performance goals); treatment of the presenting injury/illness; and treatment to correct the cause. Generally, no single form of treatment will correct the majority of SEM problems. A combination of different forms of treatment will usually give the best results.

Therefore, it is important for the clinician to be aware of the variety of treatments and to appreciate when their use may be appropriate. It is also important to develop as many treatment skills as possible or, alternatively, ensure access to others with particular skills. It is essential to evaluate the effectiveness of treatment constantly. If a particular treatment is not proving to be effective, it is important firstly to reconsider the diagnosis. If the diagnosis appears to be correct, other treatments should be considered.

## Meeting individual needs

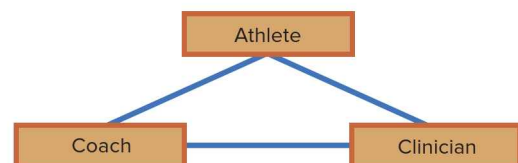
Every patient is a unique individual with specific needs. Without an understanding of this, it is not possible to manage the athlete appropriately. The patient may be an Olympic athlete whose selection depends on a peak performance at forthcoming trials. The patient may be a non-competitive business executive whose jogging is an important means of coping with the stress of everyday life. The patient may be a club tennis player whose weekly competitive game is as important as a Wimbledon final is to a professional. Alternatively, the patient may be someone to whom sport is not at all important but whose low back pain causes discomfort at work.

The cost of treatment should also be considered. Does the athlete merely require a diagnosis and reassurance that he or she has no major injury? Or does the athlete want twice-daily treatment in order to be able to play in an important game? Treatment depends on the patient's situation, not purely on the diagnosis.

## THE COACH, THE ATHLETE AND THE CLINICIAN

The relationship between the coach, the athlete and the clinician is shown in Figure 1.3. The clinician obviously needs to develop a good relationship with the athlete. A feeling of mutual trust and confidence would lead to the athlete feeling that he or she can confide in the clinician and the clinician feeling that the athlete will comply with advice.

As the coach is directly responsible for the athlete's training and performance, it is essential to involve the



**Figure 1.3** The relationship between the coach, the athlete and the clinician

coach in clinical decision making. Involving the coach in the decision-making process is essential for athlete compliance. The coach will also be a valuable aid in supervising the recommended treatment or rehabilitation program. Discussion with the coach may help to establish a possible technique-related cause for the injury. Unfortunately, some coaches have a distrust of clinicians; however, it is essential for the coach to understand that the clinician is also aiming to maximise the performance and health of the athlete. When major injuries occur, professional athletes' agents will also be involved in discussions. It is therefore crucial that clinicians understand the performance environment and develop skills in communication and management to complement optimal decision making in challenging situations.

## 'LOVE THY SPORT' (AND PHYSICAL ACTIVITY!)

To be a successful SEM clinician it is essential to be an advocate for physical activity. A good understanding of a sport confers two advantages. First, if the clinician understands the physical demands and technical aspects of a particular sport, this will improve his or her understanding of possible causes of injury and facilitate development of sport-specific rehabilitation programs. Second, it will result in the athlete having increased confidence in the clinician. The best way to understand the sport is to attend training and competition and ideally to participate in the sport. Thus, it is essential to be on site, not only to be available when injuries occur, but also to develop a thorough understanding of the sport and its culture.

### The case manager approach

**The Isokinetic group of sports medicine clinics in Italy and the United Kingdom adopt the case manager approach to rehabilitation of the injured athlete. Here the group's founder, sports physician Dr Stefano della Villa, explains the Isokinetic philosophy.**

The final result of our job, as a sport physician, is when the athlete is able to play again; as fast, as strong and as powerful as before. Everything in the middle is just a ring in the chain. Despite everyone being aware of this, in real life the situation is completely different.

Most of the time, sports physicians are unable to follow the patient during the long process from injury to return to play. A long list of practical issues may become obstacles along the way, including:

- the length of the process, especially if it is post-surgery
- the number of professionals involved, from the surgeon to the team coach and all those in the middle
- logistical issues during the recovery process—the needs of the patient during recovery to use different facilities and environments such as consultation rooms, gyms, pools and pitches, all of which may well be located in different areas
- pressure from the outside—family, friends, agent, manager, other players, coaches, fitness team, traditional media and social media.

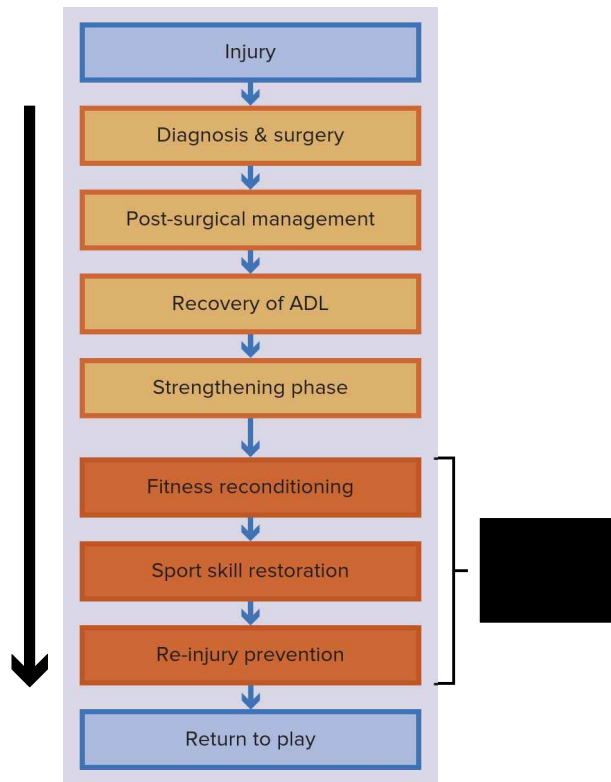
In my opinion, this long list is not the biggest obstacle to overcome. That obstacle is our way of thinking as doctors. The key is to change this paradigm: we need to forget the injury and to re-focus on the recovery from the first moment after the injury. We must do it, from the very beginning, when we explain the recovery process to the patient.

#### Medical training and its influence

Studying at university does not help the sports physician to follow the approach described above. During our time at university we learn a lot about anatomy and symptoms of diseases. The real challenge for a student is to be prepared and ready to make the right diagnosis. However, the right diagnosis does not have value without a strategy to ensure the patient returns to sport. The right diagnosis is crucial, but it is only the first step. Then the process begins and the patient goes through different steps, such as in the following example of an injury that requires surgical treatment:

- imaging
- surgical procedure
- pain management and prevention of deep venous thrombosis
- post-surgical protocols
- recovery of activities of daily living (ADL)
- recovery of range of motion
- recovery of strength
- recovery of coordination and proprioception
- recovery of match fitness
- restoration of specific sports skills
- prevention of re-injury.

We believe the sports physician should always be in charge and control all of these steps (Fig. 1.4). This, however, is not what happens in the sports medicine community. Apart from a few examples in professional sports, most of the time responsibility is shared during the recovery process, without central coordination. In the end this means that no one person is responsible. Many professionals are involved in this process, as it must be, but each of them is focused only on



**Figure 1.4** From injury to return to play

their single job and there is a need for a leader to coordinate all of them.

#### The case manager concept

We believe that the management of sports injuries can be visualised like a symphony. Many musicians are involved, but they need a conductor—a director of the orchestra. This person is able to deal with the first violin and all the other elements. Based on our experience, our medical group has defined this role as that of the ‘case manager’—the manager of each specific patient and their recovery from injury.

We strongly believe in the concept ‘patient first’ and we put the patient at the centre of our attention. We also understand that after a bad injury the patient needs a project, a plan and a conductor who is able to drive them through the potentially long recovery process.

#### The facilities for the sports recovery process

Utilising this new approach, sports physicians should always be a step ahead in the process. During the diagnosis process they must look ahead to the next possible steps. They need to have a vision of the future for the patient and understand where there could be obstacles on the road to recovery. They need to anticipate problems but, more importantly, consider

when there will be key decision points. This will include the times when the path and speed of the recovery can change, such as the options for surgery and the surgeon. Both before, and at the time of, the surgery they must plan the post-surgical management, right up to completion of the final phase. This can include seeking activities that may ordinarily appear in a future phase of rehabilitation and bringing them forward to operate in parallel with earlier phase activities. An example might be using a pool early to maintain cardiorespiratory fitness, while working to regain strength in the injured limb. Therefore, having the appropriate facilities is a first and major step in planning effective rehabilitation.

We believe for an athlete to fully complete their rehabilitation, they need access to a proper facility consisting of medical offices, rehabilitation gyms, rehabilitation pools and sports fields. The use of these specific areas at well-defined moments in the pathway is critical for the best functional outcome. For example, using a pool-based environment allows the early introduction of sport-specific movement patterns, such as kicking or heading in football (Fig. 1.5).

These exercises will be introduced again in the later phases, both in the gym and on the field, creating a type of continuum in the recovery process. It is a form of stimulation



**Figure 1.5** Neuroplasticity exercises in the pool. Heading a football with partial-body submersion allows safe football-specific movement patterns while in the early phases of rehabilitation

and is thought to favour the re-education of motor skills. In fact, after the injury there may be a process of joint de-efferentation and central disorganisation. The early introduction of some complex stimuli is recommended to solve neuromuscular impairments.

The rehabilitation gym is still considered the main rehabilitation area. Aside from the classical range of motion and strength exercises, it is useful and important to introduce more neuromuscular exercises. These exercises should not only consider the affected joint, but also the whole kinetic chain and the biomechanical connection among different joints.

According to our philosophy, we progressively introduce sport-specific movements on a real sports field, to better restore players' self-confidence in a supervised environment. The on-field rehabilitation (OFR) offers three main advantages: the complete recovery of specific movement patterns, metabolic reconditioning and a real education of the athlete in prevention strategies.

#### Teamwork and communication

With regard to the sports medicine team, we think it should at least consist of a sports medicine physician, a physical therapist and a conditioning specialist. Together they follow the player from injury to return to play.

The final key point of a good strategy is to have a shared working method within the clinical team. Within our philosophy the physician should act as the case manager. He or she is in charge of controlling the process, from the beginning to the end, communicating regularly and coordinating the team around the patient, and planning a customised protocol. Having a strong and consistent communication model within the team is critical for success. Frequent clinical updates are important to constantly monitor patient's improvements and solve eventual complications. In fact, the strength of a sports medicine group has to be measured in the management of difficult cases, when both logistical considerations and clinical skills are required.

## REFERENCES

References for this chapter can be found at [www.mhhe.com/au/CSM5e](http://www.mhhe.com/au/CSM5e)